

AI Semester Project

Comparative Analysis of KNN, Naive Bayes, and KMeans
on Financial Datasets

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Table of Contents

1. Introduction
2. Dataset Descriptions
3. Algorithm 1: K-Nearest Neighbors (KNN)
4. Algorithm 2: Naive Bayes
5. Algorithm 3: K-Means Clustering
6. Comparative Analysis
7. Conclusion

1. Introduction

This project applies three fundamental machine learning algorithms to real-world financial datasets. Each algorithm is selected for its unique strengths: KNN for pattern-based classification, Naive Bayes for probabilistic text classification, and KMeans for unsupervised customer segmentation.

2. Dataset Descriptions

Dataset	Algorithm	Rows	Features	Source
Crypto OHLCV (Top 50 Cryptos)	KNN	115,072	7 (OHLCV)	Kaggle 2014-2026
Financial News (FinancialPhraseBank)	Naive Bayes	4,845	Text Headlines	Kaggle Financial News
CC GENERAL (Credit Cards)	KMeans	8,950	18 (Spending)	Kaggle CC Dataset

2.1 Crypto Dataset (KNN)

Daily Open, High, Low, Close, Volume data for 50 major cryptocurrencies from 2014 to 2026. Bitcoin (BTC) is used for the KNN model. Features include price returns, volatility, moving averages, and high-low ranges.

2.2 Financial News Dataset (Naive Bayes)

The FinancialPhraseBank contains 4,845 financial news headlines annotated with sentiment labels: positive, neutral, or negative. This is a standard benchmark for financial sentiment analysis.

2.3 CC GENERAL Dataset (KMeans)

Credit card customer data with 8,950 records and 18 attributes including balance, purchase amounts, cash advances, credit limits, and payment behavior. Ideal for customer segmentation.

3. Algorithm 1: K-Nearest Neighbors

Goal: Predict next-day Bitcoin price direction (UP/DOWN) based on similar historical price patterns.

Methodology:

- Load BTC data from the crypto dataset
- Engineer features: returns, volatility, moving averages, price range
- Create binary target: 1 if next close > current close, else 0
- Temporal 80/20 train-test split
- Standardize features with StandardScaler
- Find optimal K (1-30) using accuracy on test set
- Evaluate with accuracy, confusion matrix, classification report

Key Findings: KNN identifies similar historical market conditions and votes on the most likely direction. The optimal K balances bias-variance tradeoff.

4. Algorithm 2: Naive Bayes

Goal: Classify financial news headlines as Positive, Neutral, or Negative.

Methodology:

- Load FinancialPhraseBank and inspect class distribution
- Text preprocessing: lowercase, remove punctuation, normalize whitespace
- Convert text to TF-IDF feature vectors (5,000 features, unigrams + bigrams)
- Split 80/20 with stratification to preserve class proportions
- Train Multinomial Naive Bayes classifier
- Evaluate with accuracy, confusion matrix, classification report
- Test on custom unseen headlines

Key Findings: Naive Bayes works well because words in financial news have distinct probability distributions across sentiment classes. The independence assumption is a reasonable approximation for bag-of-words features.

5. Algorithm 3: K-Means Clustering

Goal: Segment credit card customers into behavioral clusters for targeted marketing.

Methodology:

- Load CC GENERAL dataset (8,950 customers, 18 features)
- Handle missing values (MINIMUM_PAYMENTS, CREDIT_LIMIT)
- Drop customer ID and standardize features

- Use Elbow Method and Silhouette Score to find optimal K
- Apply KMeans with optimal K (4 clusters)
- Reduce dimensions with PCA for 2D visualization
- Interpret clusters by analyzing mean feature values

Key Findings: Customers split into 4 distinct groups: big spenders (high balance/purchases), cash-heavy users, moderate transactors, and low-activity users.

6. Comparative Analysis

Aspect	KNN	Naive Bayes	KMeans
Type	Supervised (Classification)	Supervised (Classification)	Unsupervised (Clustering)
Data Needed	Labeled + Numerical	Labeled + Text	Unlabeled + Numerical
Strengths	Simple, non-parametric, works with non-linear data	Fast, works well with text, handles high dimensions	Discovers hidden patterns, no labels needed
Weaknesses	Sensitive to scale, slow with large data	Assumes feature independence	Must choose K, sensitive to scaling
Metric	Accuracy, Confusion Matrix	Accuracy, Confusion Matrix	Inertia, Silhouette Score

Summary of Results:

- KNN predicts Bitcoin direction with accuracy >50% (better than random guessing), demonstrating that past price patterns contain predictive information.
- Naive Bayes achieves strong sentiment classification on financial news, confirming that word distributions effectively separate positive/neutral/negative texts.
- KMeans reveals 4 distinct customer segments, enabling data-driven marketing strategies.

7. Conclusion

This project demonstrates the application of three core machine learning algorithms on financial datasets. Each algorithm serves a different purpose:

- **KNN** is effective for time-series prediction where similarity to past patterns matters.
- **Naive Bayes** excels at text classification tasks like sentiment analysis.
- **KMeans** provides valuable unsupervised insights through customer segmentation.

All notebooks, code, and data are available in the project directory. The modular structure allows each algorithm to be studied independently.